

Retail Store Sales Analysis — Auto Analysis

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Executive Summary

This report summarizes the dataset quality, exploratory findings, and model performance for project "Retail Store Sales Analysis".

Dataset Overview

File	retail_sales_dataset.csv
Rows	3000
Columns	14
Missing Values	0
Duplicate Rows	0
Cleaned	True

Modeling & Evaluation

Model Type	xgboost
Problem Type	classification
Target Column	store_city

Metrics

Metric	Value
accuracy	0.265
precision	0.2319
recall	0.265
f1	0.2456
roc_auc	0.5012

confusion_matrix	{'labels': ['Faisalabad', 'Islamabad', 'Karachi', 'Lahore', 'Multan'], 'matrix': [[6, 16, 38, 20, 2], [14, 28, 45, 36, 3], [13, 35, 74, 53, 3], [14, 28, 54, 51, 6], [12, 9, 22, 18, 0]]}
class_distribution	{'Karachi': 178, 'Lahore': 153, 'Islamabad': 126, 'Faisalabad': 82, 'Multan': 61}
cv_scores	[0.22302285893341534, 0.23462340474804239, 0.24668614575498893, 0.23390032848422335, 0.2428441838327327]
cv_mean	0.2362
baseline	{'baseline_score': 0.1357, 'baseline_metric': 'f1', 'model_score': 0.2456, 'improvement': 0.1099, 'cv_std': 0.0082, 'clearly_beats_baseline': True}
leakage_warnings	[]
post_outcome_warnings	[]

Feature Importance

Feature	Importance
returned_No	0.0504
product_category_Clothing	0.0499
product_category_Electronics	0.0487
product_category_Sports	0.0472
sales_amount	0.0461
customer_rating	0.0447
promotion_applied_No	0.0443
order_date_dayofweek	0.0443
unit_price	0.0442
profit	0.044
order_date_year	0.0436
product_category_Home & Kitchen	0.0431
order_date_month	0.043

product_category_Groceries	0.043
quantity	0.0429
sales_channel_In-Store	0.0426
discount_percent	0.0424
customer_segment_Premium	0.0413
sales_channel_Online	0.0406
sales_channel_Mobile App	0.0405

AI Business Insight [AI-generated insight](#)

AI service is currently unavailable. This report contains verified analytical results only; AI-generated narrative could not be produced.

Limitations

Model predictions are estimates based on historical data and may contain uncertainty. AI-generated text explains verified analytical results but does not independently calculate metrics. Always validate significant business decisions with domain expertise.